

Topoi (ILF)

Goldblatt / Exercises / Reading group

Summer 2026

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Part I

Introduction

Meeting II

Category Theory Pt. 1

1.1 Minimal Exercises

1.1.1 Pre-order Category

For each natural number n , consider the pre-order category $\mathbf{n}$ associated with the pre-order $0 \leq 1 \leq \dots \leq n-1$. How many arrows does $\mathbf{n}$ have?

1.1.2 Subcategories of a Monoid

Consider the monoid $\mathbf{N} = (N, +, 0)$ of natural numbers, which gives rise to a category whose collection of objects is the singleton $\{N\}$ and whose collection of arrows is the set N of natural numbers. Which of the following $\mathcal{C}$ are valid subcategories?

- The empty category $\mathcal{C} = \emptyset$
- $\mathcal{C}$ -objects are $\{N\}$, and $\mathcal{C}$ -arrows are $N \setminus \{0\} = \{1, 2, 3, 4, \dots\}$
- $\mathcal{C}$ -objects are $\{N\}$, and $\mathcal{C}$ -arrows are $N \setminus \{1\} = \{0, 2, 3, 4, \dots\}$
- $\mathcal{C}$ -objects are $\{N\}$, and $\mathcal{C}$ -arrows are $N \setminus \{2\} = \{0, 1, 3, 4, \dots\}$

1.2 Exercises for Good Preparation

1.2.1 Endomorphism Monoid

Let $\mathcal{C}$ be a category and X be an object of $\mathcal{C}$. Verify that the collection $\mathcal{C}(X, X)$ of all $\mathcal{C}$ -arrows from X to X forms a monoid with respect to composition.

1.2.2 Sum Category

Let $\mathcal{C}$ and $\mathcal{D}$ be categories. The sum category, denoted $\mathcal{C} + \mathcal{D}$, is a category whose collection of objects is the disjoint union of $\mathcal{C}$ -objects and $\mathcal{D}$ -objects, and whose collection of arrows is the disjoint union of $\mathcal{C}$ -arrows and $\mathcal{D}$ -arrows.

- Describe how to inherit domains, codomains, identity, and composition from both $\mathcal{C}$ and $\mathcal{D}$. Verify that $\mathcal{C} + \mathcal{D}$ is a category.
- Verify that there are no $(\mathcal{C} + \mathcal{D})$ -arrows between any object that came from $\mathcal{C}$ and any object

that came from $\mathcal{D}$.

- Find an example of $\mathcal{C}$ and $\mathcal{D}$ where if set union was used instead of disjoint union, then the resulting construction would not yield a category.

1.3 Extra Exercise

1.3.1 Quivers

A *quiver* is a directed graph that allows loops and multiple parallel edges. In particular, a quiver Q comprises:

- a collection of things called *Q-vertices*;
- a collection of things called *Q-edges*;
- operations assigning to each *Q-edge* f a *Q-vertex* $s(f)$ (the “source” of f) and a *Q-vertex* $t(f)$ (the “target” of f).

Every category has an underlying quiver by forgetting the composition (and identity): each object is a vertex, each arrow is an edge, $s(f) = \text{dom}(f)$, and $t(f) = \text{cod}(f)$.

Two quivers are considered isomorphic when there is a bijection of vertices and a bijection of edges that preserves sources and targets. Two categories are considered isomorphic when there exists an isomorphism of the underlying quivers that additionally preserves composition and identity.

Find an example of two non-isomorphic categories such that the underlying quivers are isomorphic.

Meeting III

Category Theory Pt. 2

2.1 Minimal Exercises

2.1.1 Terminal Objects

Selected exercises from Goldblatt section 3.6:

1. For any category $\mathcal{C}$, prove that all terminal $\mathcal{C}$ -objects (if any) are isomorphic (to each other).
2. Find terminals in $\mathbf{Set}^2$, $\mathbf{Set}^\rightarrow$, and the poset $\mathbf{n}$.

2.1.2 Products

Selected exercises from Goldblatt section 3.8:

Let a, b be two $\mathcal{C}$ -objects that have a product $a \times b$ with projections $\text{pr}_a : a \times b \rightarrow a$ and $\text{pr}_b : a \times b \rightarrow b$.

Exercise 1: Prove that $\langle \text{pr}_a, \text{pr}_b \rangle = 1_{a \times b}$.

Exercise 2: Prove that if $\langle f, g \rangle = \langle k, h \rangle$, then $f = k$ and $g = h$.

Exercise 3: Prove that $\langle f \circ h, g \circ h \rangle = \langle f, g \rangle \circ h$.

Exercise 4: Show that if $\mathcal{C}$ has a terminal object and products, then for any $\mathcal{C}$ -object a , $a \cong a \times 1$ and indeed $\langle 1_a, l_a \rangle$ is iso.

Exercise 5: $1_a \times 1_b = 1_{a \times b}$

Exercise 6: $a \times b \cong b \times a$

2.2 Exercises for Good Preparation

2.2.1 Monic Arrows

Exercises from Goldblatt section 3.1: in any category, consider arrows $a \xrightarrow{f} b$, $b \xrightarrow{g} c$ as in the commutative diagram:

$$\begin{array}{ccc} a & \xrightarrow{f} & b \\ & \searrow g \circ f & \downarrow g \\ & & c \end{array}$$

- (1) Prove that if f and g are both monic, then f is monic.
- (2) Prove that if $g \circ f$ is monic, then f is monic.

2.2.2 Iso Arrows

Exercises from Goldblatt section 3.3: in any category, verify the following:

1. Every identity arrow is iso.
2. If $b \xrightarrow{f} c$ is iso, so is $c \xrightarrow{f^{-1}} b$.
3. $a \xrightarrow{f \circ g} c$ is iso if $b \xrightarrow{f} c$, $a \xrightarrow{g} b$ are, with $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$.

2.2.3 Terminal Objects

Selected exercises from Goldblatt section 3.6:

3. Show that an arrow $1 \rightarrow a$ whose domain is a terminal object must be monic.

2.2.4 Isomorphic Objects

Exercise 1 of 2 from Goldblatt section 3.4: For any $\mathcal{C}$ -objects a, b, c in any category $\mathcal{C}$, prove the following:

- (i) $a \cong a$
- (ii) if $a \cong b$, then $b \cong a$
- (iii) if $a \cong b$ and $b \cong c$, then $a \cong c$

2.3 Extra Exercises